

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary - April 2023

20EC305 - SIGNALS AND SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit - I

Marks BT* CO* PO*

1. a) The trapezoidal shown in Fig. Q1(a) is applied to a differentiator whose

$$\text{output is } y(t) = \frac{dx(t)}{dt}$$

- Sketch the differentiator output
- Find energy of the differentiator output

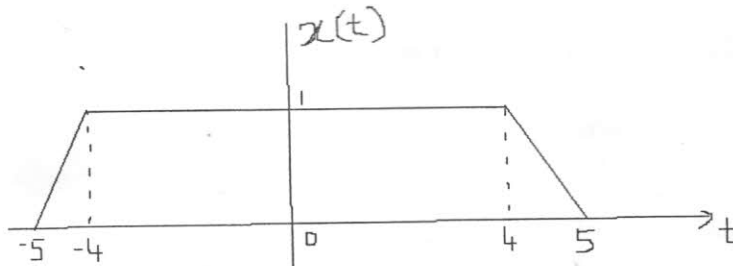


Fig. Q1(a)

- b) Find whether or not each of the following signal is periodic or not. If periodic, find its fundamental period.

(i) $x(t) = \cos(5\pi t) + \sin(6\pi t)$

(ii) $x(t) = \cos(\sqrt{2}t) + \cos t$

- c) Plot the signal

$$y(n) = x(n)u(2-n)$$

Where $x(n) = \{1, 1, 1, 1, 1, 0.5\}$

2. a) Sketch the waveforms for the following signals

(i) $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$

(ii) $x(t) = r(t+2) - r(t+1) - r(t-1) + r(t-2)$

(iii) $x(n) = [6-n][u(n) - u(n-6)]$

(iv) $x(n) = 2^n[u(n+1) - u(n-4)]$

- b) Sketch even & odd parts of the signal $u(n)$.

- c) Find energy or power of the following signals.

(i) $x(t) = A\cos(\omega_0 t + \phi)$

(ii) $x(n) = u(n)$

Unit - II

3. a) Check the system represented by the following equation for linearity, causality, time invariance and stability.

(i) $x(-n)$ (ii) $\frac{dx(t)}{dt}$

- b) Find the output of the system and also sketch the same.

$$x(t) = e^{-at}u(t); a > 0$$

$$h(t) = u(t)$$

- c) State and prove associative property of convolution sum. 6 L2 2 1
4. a) Show that (i) $x(t) * \delta(t) = x(t)$ (ii) $x(t) * \delta(t - t_0) = x(t - t_0)$
 (iii) $x(t) * u(t - t_0) = \int_{-\infty}^{t-t_0} x(\tau) d\tau$ 8 L3 2 1
 b) Convolve the following signals; $\beta^n u(n)$ and $u(n - 3)$; $|\beta| < 1$ 4 L3 2 1
 c) State and Prove commutative and distributive property of convolution Integral. 8 L2 2 1

Unit – III

5. a) The impulse response of LTI systems are given below. Find whether each system is memory less, causal and stable.
 (i) $h(t) = e^{-3t} u(t - 1)$ (ii) $h(n) = (1/2)^n u(n)$ 8 L3 3 1
 b) Realize the difference equation in direct form I and direct form II methods.
 $y(n] + 0.25y(n - 1) - 0.125y(n - 2) = x(n) + x(n - 1)$ 4 L3 3 1
 c) Find the zero-state response for the system described by the difference equation
 $y(n] - \frac{5}{6}y(n - 1) + \frac{1}{6}y(n - 2) = x(n)$; where $x(n) = 2^n u(n)$ 8 L3 3 1
6. a) The impulse response of LTI systems are given below. Find whether each system is memory less, causal and stable.
 (i) $h(t) = 3\delta(t)$ (ii) $h(n) = 2u(n) - 2u(n - 1)$ 8 L3 3 1
 b) Realize the differential equation in direct form I and direct form II methods.
 $\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 4y(t) = \frac{dx(t)}{dt}$ 6 L3 3 1
 c) Prove that a Continuous LTI System is stable only if its impulse response is absolutely integrable. 6 L2 3 1

Unit – IV

7. a) Evaluate the Fourier Series representation for the signal
 $x(t) = \cos(4t) + \sin(8t)$
 Also sketch the magnitude and phase spectra 6 L3 4 1
 b) State and Prove frequency shift property of CTFS. 6 L2 4 1
 c) Evaluate the DTFS representation for the signal
 $x(n) = \cos\left(\frac{\pi}{3}n\right)$
 Also sketch the spectrum. 8 L3 4 1
8. a) Find the time domain signal whose FS coefficient is given by
 $X(K) = \cos\left(\frac{10\pi}{21}K\right) + j\sin\left(\frac{4\pi}{21}K\right)$ 4 L3 4 1
 b) State and Prove convolution and Parseval's relation properties of DTFS 10 L2 4 1
 c) Find the time domain signal corresponding to the DTFS coefficients $X(K)$ and sketch it.
 $X(K) = \cos\left(\frac{6\pi}{17}K\right)$ 6 L3 4 1

Unit – V

9. a) Find the DTFT of unit step sequence. 6 L3 4 1
 b) Obtain the Fourier transform of the signal
 $x(t) = e^{-at} u(t)$; $a > 0$
 Also. draw its magnitude and phase spectra. 8 L3 4 1

- c) Find the DTFT of the signal $x(n) = \{1, 3, 5, 3, 1\}$ and evaluate $X(e^{j\Omega})$ at $\Omega = 0$

6 L3 4 1

10. a) Find the Fourier Transform of $x(t) = e^{-a|t|}$; $a > 0$. Draw its spectrum.

6 L3 4 1

- b) Specify the Nyquist rate for each of the following signals.

(i) $x_1(t) = \text{sinc}(200t)$ (ii) $x_2(t) = \text{sinc}^2(200t)$

6 L3 4 1

- c) Obtain the frequency response & the impulse response of the system described by the difference equation

$$y(n) + \frac{1}{2}y(n-1) = x(n) - 2x(n-1)$$

8 L3 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
